

Venue Activity Prediction: a model for Social Media Integration

(working title)

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Abstract

Placeholder. Write the abstract last, once the model and the simulation results exist.

1 Introduction

2 Literature Review

3 Problem Statement

A social app lets users post a check-in at a public venue. A check-in records one drink, so one visit can produce several check-ins. Each check-in carries a user, a timestamp, and a venue. The app accepts a check-in only at the venue's own location. It serves a community of young adults who want to socialize with friends. We treat the community as a finite set, fixed only at the moment the app judges it, and its size can differ from one night to the next.

We distinguish three objects. The community network carries the friendships that decide who goes out together. The adopter set carries the members who use the app. The app's own graph carries the follows it has recorded. Only the third is observable.

Definition 1 (The three layers). *Let $C = (V, E)$ be the community network, where V is the community and E holds the mutual friendships among its members. Let $A \subseteq V$ be the adopter set, the members who use the app, and call $|A|/|V|$ the adoption rate. Let $G = (A, F)$ be the app graph, where $F \subseteq \{e \in E : e \subseteq A\}$ holds the mutual follows the app has recorded. The app observes G . It observes neither C nor $V \setminus A$.*

Assumption 1 (No non-adopter signal). *Members of $V \setminus A$ post nothing.*

Assumption 2 (Adopter- and venue-dependent posting probability). *A present adopter posts at least once with positive probability. That probability can depend on the adopter and on the venue.*

Assumption 3 (Friends at the same venue and time). *Non-friends post independently of one another. Friends at the same venue and time do not.*

Assumption 3 quantifies dependence over E , not over F . The app therefore faces correlation whose source it need not have recorded. Assumption 1 bounds the consequence. Correlated posting arises only between two co-present adopters joined in E , since non-adopters post nothing and non-friends post independently. The following assumption makes that bound exact.

Assumption 4 (Information-complete app graph). $F = \{e \in E : e \subseteq A\}$.

Assumption 4 does not claim that the app records friendships perfectly. It states the case in which the app holds every tie capable of producing a correlated post. We take this case first, because it bounds from above what any procedure using the app graph can achieve. We treat $F \subsetneq \{e \in E : e \subseteq A\}$ as a later relaxation.

Definition 2 (True headcount and occupancy). *Each venue holds a definite number of community members at any moment. We call that number its true headcount, written N_v . Each venue has a fixed capacity K_v , and we call N_v/K_v its true occupancy. The app reports neither quantity.*

The app judges each venue from check-ins posted there in the last two hours, showing every venue with at least one such check-in. Each shown venue gets one of three levels: quiet, busy, or packed. The two boundaries are the terciles of tonight’s occupancy distribution among open venues, not a fixed historical threshold. Users compare levels, not numbers.

Definition 3 (Match rate and coverage). *The app gets a shown venue right when its level matches its true occupancy. The fraction of shown venues it gets right is the match rate. A venue that never appears is a different kind of error. The fraction of busy-or-packed venues that appear at all is the coverage.*

Coverage carries no reference to a procedure. A venue appears when at least one adopter present posts, whatever the app then computes from the posts. Coverage therefore depends on C , on A , and on the posting probabilities, and not on G .

Definition 4 (The question). *Fix a coverage target $\tau \in (0, 1)$. We ask which placements of A within C achieve coverage at least τ .*

We vary structure rather than adoption rate. Two quantities govern the answer: the size of A , and the degree to which A concentrates on C . A concentrated A occupies few friend groups. Those groups fill few venues, and the remaining venues emit nothing. A dispersed A of the same size meets many groups and misses little. The target τ therefore determines a curve in those two quantities, not a single adoption rate.

The app graph serves the levels rather than the coverage. It distinguishes ten check-ins from one group of friends from ten check-ins by ten strangers, which lets a procedure discount correlated posts instead of reading them as independent evidence. It also carries information about the coverage itself, since a dense G indicates that A concentrates on C . Its value for the levels grows with that concentration, because concentration is what generates the correlation it corrects.

We answer the question inside a synthetic simulation. The simulation generates a venue map, a flow of simulated people, and a social graph, and it fixes N_v by construction. We use no real check-in data and no real venue data. We may set the generator’s parameters from published summary statistics, which leaves the graph generated rather than observed.

We compare coverage against the coverage that a uniformly placed adopter set of the same size achieves. We compare a level assignment against three reference points, fixed before we read any result: chance, computed rather than assumed; the rule an operator would otherwise ship, meaning venues ordered by check-in count, by distinct posters, or by count over capacity; and the optimal rule for this loss, which requires N_v and therefore exists only inside the simulation.

Three limits bound what follows. We report simulation evidence and not a theorem, and the simulator is our own construction. The concentration of A on C has no external anchor, since no source we examined reports such a figure for an adopter set on a friendship network, so we choose the range we sweep. And Fithian, Elith, Hastie and Keith report that thinning leaves relative intensity unrecoverable when a thinning covariate coincides with an intensity covariate. Assumption 2 lets posting depend on the venue while occupancy is indexed by the venue, so the two margins of this problem share a covariate. We treat that result as a soundness check on our level assignments rather than as a settled objection, because the step transferring it to this setting is unproved.

4	Model
5	Simulation
6	Results
7	Discussion
	References